

Documentation

Introduction & Topic

For this project, our group opted to construct a **cone** with maximum radius 32" and height, from the xy-plane, 36" (or 2.5" and 3", respectively). The cone was limited to only positive z values, for the sake of construction, and no other domain restrictions were imposed.

The official equation used was

$$\frac{x^2}{900} + \frac{y^2}{900} = \frac{z^2}{1296}, \quad z > 0,$$

and we ...

Construction Process

...

Calculations

Precise Theoretic Value

To calculate the precise theoretic value, we will use an integral, specifically a double integral.

Approximate Theoretic Value

To calculate the approximate theoretic value, we will use a Riemann summation, specifically two summations.

Discussion

Precise Theoretic Value

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Approximate Theoretic Value

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